

Reinforcement Learning

Monte Carlo Methods Sutton/Barto* Chapter 5

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With figures from Sutton/Barto*

*Sutton and Barto, Reinforcement Learning: An Introduction,
2nd edition, MIT Press, Cambridge, MA, 2018



Summary of Notation

General

X	capital letters: random variables
x, p	lower-case letters: realizations of random variables or scalar functions
$\mathbf{w}$	Bold lower-case letters: real-valued vectors (even if random variables)
$\mathbf{W}$	bold capitals: matrices
α	Greek letters: parameters (vectors if in bolt)
$\Pr\{X = x\}$	probability that a random variable X takes on the value x
$X \sim p$	random variable X selected from distribution $p(x) = \Pr\{X = x\}$
$\mathbb{E}[X]$	expectation of a random variable X , i.e., $\mathbb{E}[X] = \sum_x p(x)x$
$\operatorname{argmax}_a f(a)$	a value of action a at which $f(a)$ takes its maximal value

Value Function

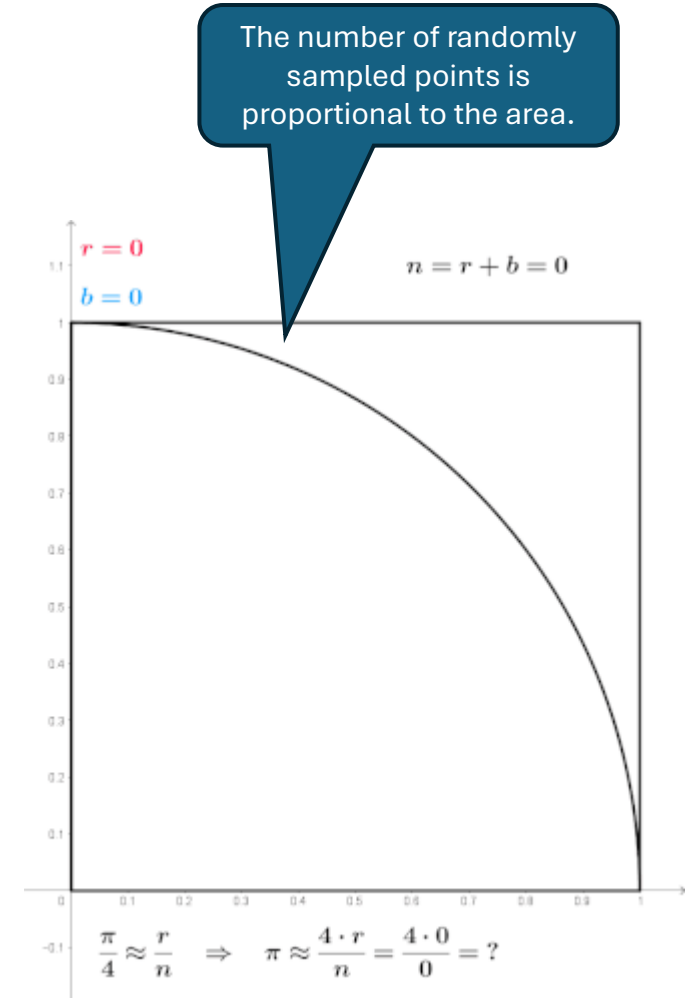
G_t	return (cumulative reward) following time t
$G_{t:h}$	return from t to h (discounted and corrected)
$v_\pi(s)$	value of state s under policy π (expected return)
$v_*(s)$	value of state s under the optimal policy
V, V_t	array estimates of state-value function v_π or v_*
$q_\pi(s, a)$	value of taking action a in state s under policy π
$q_*(s, a)$	value of taking action a in state s under the optimal policy
Q, Q_t	array estimates of action-value function q_π or q_*

MDP

s, s'	states
a	an action
r	a reward
$\mathcal{S}$	set of all (nonterminal) states, $\mathcal{S}^+$ are all states
$\mathcal{A}(s)$	set of all actions available in state s
γ	discount-rate parameter
t	discrete time step
T	final time step of an episode (a.k.a. horizon)
A_t	random variable for the action at time t
S_t	random variable for the state at time t
R_t	random variable for the reward at time t
$p(s', r s, a)$	probability of transition to state s' and receiving reward r , from state s taking action a
	expected immediate reward from state s after action a
$\pi(a s)$	probability of taking action a in state s under stochastic policy π
$\pi(s)$	action taken in state s under deterministic policy π

Introduction

- No complete knowledge of the environment (the MDP). This is called a “**model-free**” method.
- We use “experience” = **sample sequences** of states, actions, and rewards from actual or simulated interactions with the environment.
- **Advantages:**
 - We often have **sample access** to systems.
 - It is easier to simulate transitions than to specify the complete MDP for DP.
- **Monte Carlo (MC) methods**
 - a family of algorithms that rely on repeated random sampling to obtain numerical results.
 - **Applied to value function approximation:** average sample returns for each state-action pair obtained from sampled sequences (called episodes or trajectories).
- **Note:** Value estimates or the policy are only updated after a whole episode is sampled. This is not a step-by-step online method!



Monte Carlo method applied to approximating the value of π

Source: Wikipedia

On-Policy Monte Carlo Prediction

Estimate state values by sampling with a given policy.

MC Prediction: Estimate v_π

- **Goal:** Learn $v_\pi(s)$ for a given deterministic or stochastic π by averaging the observed returns G for the rest of the episode after a visit of s .
- **Sample-average method:** The average converges to $v_\pi(s)$.
Reason: state value is defined as an expected value which is approximated by the observed average.
- **First-visit:** Using only the first visit of a state in the episode ensures that the sample returns are i.i.d. distributed and we get an **unbiased** estimate.

First-visit MC prediction, for estimating $V \approx v_\pi$

Input: a policy π to be evaluated

Initialize:

$V(s) \in \mathbb{R}$, arbitrarily, for all $s \in \mathcal{S}$

$Returns(s) \leftarrow$ an empty list, for all $s \in \mathcal{S}$

Loop forever (for each episode):

Generate an episode following π : $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless S_t appears in $S_0, S_1, \dots, S_{t-1}$:

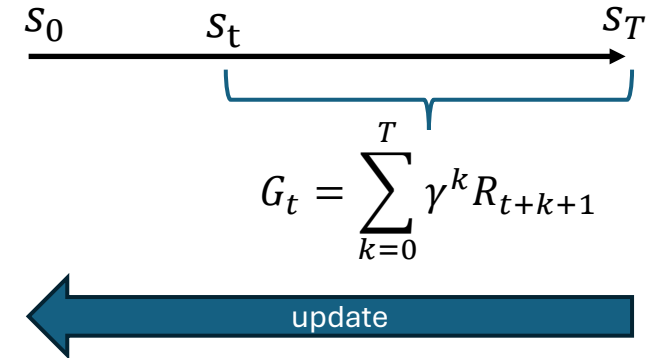
Append G to $Returns(S_t)$

$V(S_t) \leftarrow \text{average}(Returns(S_t))$

Update in reverse order!

Uses only the first visit.

Episode



On-Policy Monte Carlo Control

Find a good policy while using it to sample data.

MC Control: Find a Good Policy

- **Goal:** Find a good policy $\pi \approx \pi^*$.
- **Issue:** The state values v is not sufficient to pick the best action. We need the Q-function.

$$\pi(s) = \operatorname{argmax}_a q(s, a) = \operatorname{argmax}_a \sum_{s', r} p(s', r | s, a) [r + \gamma v(s')]$$


model p is not accessible!

- **Solution:** We need to estimate the Q-function from samples.

MC Prediction to Estimate Q-Values

- $q_{\pi}(s, a)$ can be estimated with regular MC prediction. Just record and average returns for action-state pairs (a, s) .
- **Guarantee:** Converges to q_{π} after ∞ episodes.
- **Complication:** Many (a, s) -pairs may never be visited when following π . We need to keep exploring!
- Options:
 - a) **Exploring starts:** choose the starting (a, s) for each episode randomly.
 - b) Only consider **soft policies** like ϵ -greedy.
Remember: Soft policies have non-zero probabilities for all actions so they will explore.

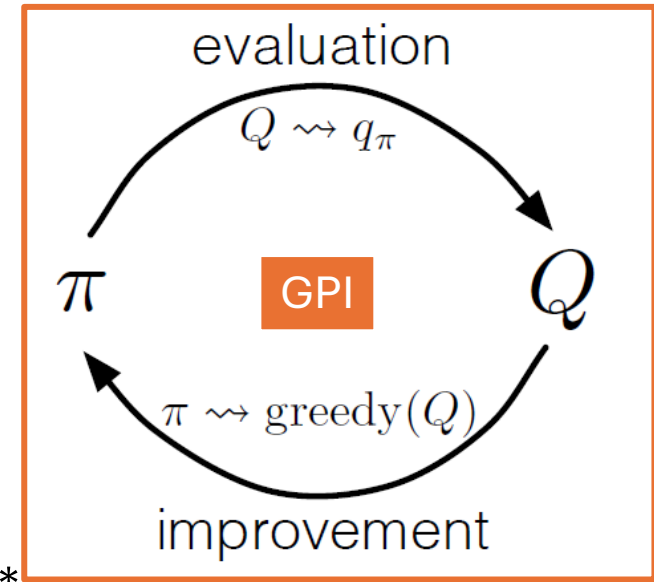
MC Control as GPI

- Approx. π_* using ideas from GPI with a Q-function.

- MC policy iteration:

$$\pi_0 \xrightarrow{E} q_{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} q_{\pi_1} \xrightarrow{I} \pi_2 \xrightarrow{E} q_{\pi_2} \xrightarrow{I} \dots \pi_* \xrightarrow{E} q_*$$

1. Start with an arbitrary policy.
2. **Evaluation:** Update MC estimate of by sampling an episode using the current π .
3. **Improvement:** greedy(Q) means $\pi(s) = \operatorname{argmax}_a q(s, a)$
4. Repeat evaluation and improvement till the optimal policy is reached.



Remember: Policy Improvement Theorem

π_{k+1} is better (equal) to π_k

$$\begin{aligned} q_{\pi_k}(s, \pi_{k+1}(s)) &= q_{\pi_k}(s, \operatorname{argmax}_a q_{\pi_k}(s, a)) \\ &= \max_a q_{\pi_k}(s, a) \\ &\geq q_{\pi_k}(s, \pi_k(s)) \\ &\geq v_{\pi_k}(s) \end{aligned}$$

→ converges to π_*

- Assumptions:
 - All states are reached (e.g., start episodes with exploring states)
 - Perfect policy evaluation with ∞ many episodes.
- Options for imperfect policy evaluation:
 - Get bounds for the error and use sufficiently many episodes to reach an acceptable level of error.
 - Instead of complete policy evaluation just move a little closer to q_{π_k} .

Alg. MC Control with Exploring Starts

First-visit MC prediction, for estimating $V \approx v_\pi$

Input: a policy π to be evaluated

Initialize:

$V(s) \in \mathbb{R}$, arbitrarily, for all $s \in \mathcal{S}$

$Returns(s) \leftarrow$ an empty list, for all $s \in \mathcal{S}$

Loop forever (for each episode):

Generate an episode following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode:

$G \leftarrow \gamma G + R_{t+1}$

Unless S_t appears in $S_0, S_1, \dots, S_{t-1}$:

Append G to $Returns(S_t)$

$V(S_t) \leftarrow \text{average}(Returns(S_t))$

Monte Carlo ES (Exploring Starts), for estimating $\pi \approx \pi_*$

Initialize:

$\pi(s) \in \mathcal{A}(s)$ (arbitrarily), for all $s \in \mathcal{S}$

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

Loop forever (for each episode):

Choose $S_0 \in \mathcal{S}, A_0 \in \mathcal{A}(S_0)$ randomly such that all pairs have probability > 0

Generate an episode from S_0, A_0 , following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \dots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$

Q-Values
instead of V .

Exploring starts

Update policy

MC Control without Exploring Starts

- **Issue:** The environment needs to let us start episodes at different (s, a) at will. This is fine for simulations, but hard in the real world.
- **Need:** We need a guarantee that all (reachable) state-action combinations are tried.
- Options:
 - a) **On-policy control:** learn a soft policy
 - b) **Off-policy control:** use a soft behavior policy for sampling that tries all state-action pairs, but learn the optimal, deterministic policy.

Types of Soft Policies

- **Soft policy:**

$$\pi(a|s) > 0 \quad \forall s \in \mathcal{S}, a \in \mathcal{A}$$

+ gradually shifts closer to the optimal deterministic policy.

- **ϵ -soft policy:**

$$\pi(a|s) \geq \frac{\epsilon}{|\mathcal{A}(s)|}$$

- **ϵ -greedy policy:** follows a greedy policy but chooses a random action with probability ϵ

$$\pi(a|s) = \begin{cases} 1 - \epsilon + \frac{\epsilon}{|\mathcal{A}(s)|} & \text{if } a = \text{greedy action} \\ \frac{\epsilon}{|\mathcal{A}(s)|} & \text{if } a \neq \text{greedy action} \end{cases}$$

Alg. On-Policy MC Control with ϵ -soft Policies

Monte Carlo ES (Exploring Starts), for estimating $\pi \approx \pi_*$

Initialize:

$\pi(s) \in \mathcal{A}(s)$ (arbitrarily), for all $s \in \mathcal{S}$

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

Loop forever (for each episode):

Choose $S_0 \in \mathcal{S}, A_0 \in \mathcal{A}(S_0)$ randomly such that

Generate an episode from S_0, A_0 , following π

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, \dots, S_{t+1}, A_{t+1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$\pi(S_t) \leftarrow \text{argmax}_a Q(S_t, a)$

On-policy first-visit MC control (for ϵ -soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\epsilon > 0$

Initialize:

$\pi \leftarrow$ an arbitrary ϵ -soft policy

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

Repeat forever (for each episode):

Generate an episode following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \dots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$A^* \leftarrow \text{argmax}_a Q(S_t, a)$

(with ties broken arbitrarily)

For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \epsilon + \epsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \epsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

Create ϵ -greedy policy.

Start with soft policy

No exploring starts!

Off-Policy Prediction

Estimate a value function for a policy different from the sampling policy.

Off-Policy vs. On-Policy Learning

Dilemma:

- We need behavior that explores. These exploratory actions will often lead to suboptimal outcomes.
- Learning action values depends on observing subsequent optimal behavior.

On-policy: Learns a near-optimal policy that still explores (i.e., ϵ -greedy with a small ϵ).

Off-policy: Use two policies

- Target policy: learn π , which will become an optimal deterministic policy.
 - Behavior policy b : guarantees exploration with a soft policy.
-
- Called “off-policy” because the generated data follows a different policy than the target policy.
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- Off-policy methods are typically:
 - More complicated
 - Converge slower
 - Often more powerful
 - Can be applied to data externally generated using a fixed, potentially unknown, behavior policy.

Note: On-policy learning is a special case of off-policy learning with $b = \pi$

Off-Policy Prediction

- **Goal:** Estimate v_π or q_π given episodes generated with $b \neq \pi$.
- **Assumptions:**
 - π and b are fixed
 - **Coverage:** Every action taken under π will occasionally also be taken under b
$$\pi(a|s) > 0 \Rightarrow b(a|s) > 0$$
 - b needs to be stochastic for states where it differs from π (typically b is ϵ -greedy)
- We are typically interested in the optimal policy π with respect to the current estimate of q . We know this policy has to be a **deterministic greedy policy**.

Importance Sampling

- **Importance sampling** is a method to estimate the expected values under a desired distribution given samples from another distribution
 - Desired distribution: actions are chosen according to the target policy π
 - Sampled distribution: actions are chosen by b
- **Use in off-policy prediction:** weight returns observed under b to estimate returns for π using the probability of trajectories under π .
- Probability of a trajectory starting at S_t under π

$$\Pr\{A_t, S_{t+1}, A_{t+1}, \dots, A_{T-1}, S_T | S_t, A_{t:T-1} \sim \pi\} = \prod_{k=t}^{T-1} \pi(A_k | S_k) p(S_{k+1} | S_k, A_k)$$

Needs unknown transition probabilities!

- **Importance sampling ratio** (probability that b chooses the same actions as π):

$$\rho_{t:T-1} \stackrel{\text{def}}{=} \frac{\prod_{k=t}^{T-1} \pi(A_k | S_k) p(S_{k+1} | S_k, A_k)}{\prod_{k=t}^{T-1} b(A_k | S_k) p(S_{k+1} | S_k, A_k)} = \prod_{k=t}^{T-1} \frac{\pi(A_k | S_k)}{b(A_k | S_k)}$$

Luckily, the unknown transition probabilities cancel out!

$$\mathbb{E}[\rho_{t:T-1} G_t | S_t = s] = v_\pi(s)$$

All we need to do is correct the returns by multiplying them with the ratio.

Note: The ratio has to be calculated for the remainder of each sampled trajectory.

Ordinary vs. Weighted Importance Sampling

- Ordinary importance sampling

$$V(s) \stackrel{\text{def}}{=} \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|}$$

$\mathcal{T}(s)$ time steps that s was visited

- Gives an unbiased estimate of v_π but variance is extreme for low $|\mathcal{T}(s)|$

- Weighted importance sampling

$$V(s) \stackrel{\text{def}}{=} \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1}}$$

- Set 0 if denominator is 0
- Initially biased to v_b . Bias falls asymptotically with more visits.
- Preferred method with much lower errors at lower number of episodes.

Incremental Implementation using Weighted Importance Sampling

On-policy first-visit MC control (for ε -soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$

Initialize:

$\pi \leftarrow$ an arbitrary ε -soft policy

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all s, a

$Returns(s, a) \leftarrow$ empty list, for all s, a

Repeat forever (for each episode $i = 1, 2, \dots$):

Generate an episode following π :

$G \leftarrow 0$

Loop for each step of episode, $t = 0, 1, \dots, T-1$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $Returns(S_t, A_t)$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$A^* \leftarrow \arg\max_a Q(S_t, a)$

For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon / |\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon / |\mathcal{A}(S_t)| & \text{otherwise} \end{cases}$$

Off-policy MC prediction (policy evaluation) for estimating $Q \approx q_\pi$

Input: an arbitrary target policy π

Initialize, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$:

$Q(s, a) \in \mathbb{R}$ (arbitrarily)

$C(s, a) \leftarrow 0$

cumulative sum of weights for (s, a) for incremental update

Loop forever (for each episode):

$b \leftarrow$ any policy with coverage of π

b can change every episode!

Generate an episode following b : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

$W \leftarrow 1$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$, while $W \neq 0$:

$G \leftarrow \gamma G + R_{t+1}$

$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$

$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$

$W \leftarrow W \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$

Incremental update + Correct the update

Off-Policy Control

Find a good policy while sampling with a different behavior policy.

Off-Policy MC Control

- **Remember:** on-policy control tries to estimate a policy while using it for control. The behavior changes with the learned policy.
- Off-policy control uses two policies:
 - Behavior policy b : needs to be soft (have coverage)
 - Target policy π : can be deterministic (e.g., greedy)

Alg. Off-Policy MC Control

Off-policy MC prediction (policy evaluation) for estimating $Q \approx q_\pi$

Input: an arbitrary target policy π

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:

$Q(s, a) \in \mathbb{R}$ (arbitrarily)

$C(s, a) \leftarrow 0$

Loop forever (for each episode)

$b \leftarrow$ any policy with coverage

Generate an episode following b :

$G \leftarrow 0$

$W \leftarrow 1$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$

$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$

$W \leftarrow W \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$

Only uses the end of the episode that uses greedy actions for π

Issue: Not very sample efficient!

Off-policy MC control, for estimating $\pi \approx \pi_*$

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:

$Q(s, a) \in \mathbb{R}$ (arbitrarily)

$C(s, a) \leftarrow 0$

$\pi(s) \leftarrow \operatorname{argmax}_a Q(s, a)$ (with ties broken consistently)

Add the target policy

Loop forever (for each episode):

$b \leftarrow$ any soft policy

Generate an episode using b : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

$W \leftarrow 1$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$

$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$

$\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)

If $A_t \neq \pi(S_t)$ then exit inner Loop (proceed to next episode)

$W \leftarrow W \frac{1}{b(A_t|S_t)}$

We use π to select a . So the probability is 1.

What you Should Know

- MC works with sample episodes:
 - **Policy Evaluation:** Averages many sampled returns.
 - **Control:** Interleaves pol. eval. and pol. improvement on an episode-by-episode basis.
- Advantage of MC methods vs DP:
 - **Model-free:** No need for an environment model. Sampling from a simulation is often easy.
 - Can evaluate a small **state set** of interest. DP always has to sweep over all states.
 - **Produced unbiased estimates:** Estimates are based on returns from complete sampled sequences. DP and other methods use bootstrapping (estimating a value from other estimates), which introduces bias.
 - Using complete sequences may be less harmed by violations of the Markov property.
- Issues:
 - **Needs to maintain sufficient exploration.** Options are:
 1. Exploring starts can learn the optimal policy.
 2. Learning an ϵ -soft policy.
 3. Off-policy learning with a covering behavior policy. Note: only a part of the data can be used.
 - MC methods are **not sample efficient**. I.e., sampled episodes are used only once and then discarded.
 - **Not a truly online method:** Updates are not done after each action, but are delayed until the end of each episode.